# Staff & Contact

**Lectures**

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**Practical and Applications**

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## Microsoft Teams

## Biostatistics and Computer 21 - 22

# SPSS

- Statistical Package for the Social Sciences

- Data file extension (*.sav)
- Output file extension (*.spv)

## Statistics

A branch of mathematics that transforms **data** into useful **information** for decision makers.

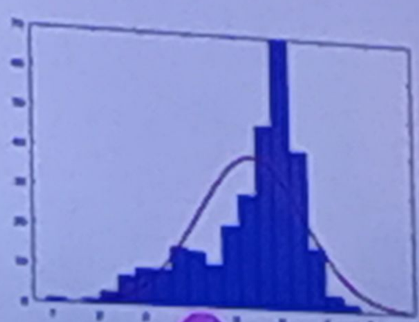
**Biostatistics** is the application of statistical methods in **biology**.

“Why we should study statistics?”

1. All fields deal with data, so all need statistics.
2. To understand statistical outputs

**As a veterinarian you may read:**

- The shelf life of an unrefrigerated egg is **8 ± 2 days.**
- Antibiotic supplement improved growth rate than probiotic supplement (**$P<0.05$**)
- **Random sample** !!!



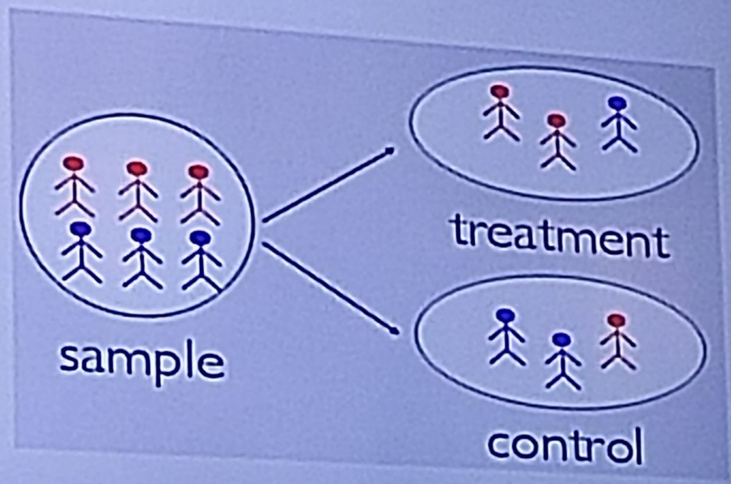
Dr. Mahmoud MA Elmoghreby

“ Why we should study statistics? ”

# Branches of Statistics

1. Experimental Design: the planning and carrying out of a study.

*(Outside the scope of our course)*



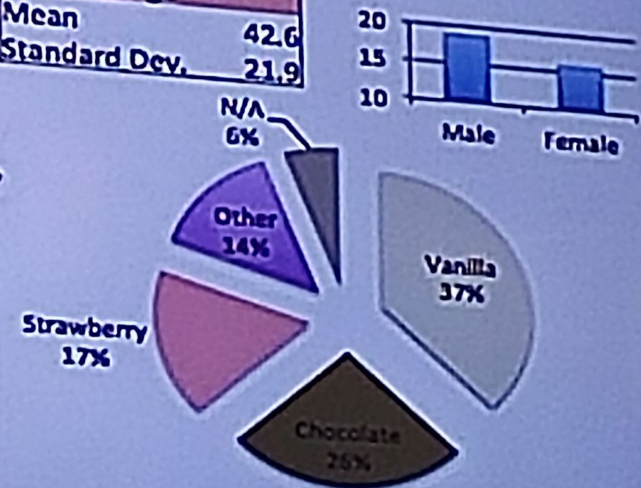
# Branches of Statistics

2. Descriptive statistics: Graphical and numerical methods used for summarizing and presenting data.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Respondent # | Age | Gender | Favorite Ice Cream Flavor |
| 2 | 1 | 36 | m | Vanilla |
| 3 | 2 | 22 | f | Chocolate |
| 4 | 3 | 61 | m | Strawberry |
| 5 | 4 | 88 | m | Other |
| 6 | 5 | 31 | m | N/A |
| 7 | 6 | 53 | m | N/A |
| 8 | 7 | 30 | f | Chocolate |
| 9 | 8 | 64 | f | Chocolate |
| 10 | 9 | 18 | m | Vanilla |
| 11 | 10 | 16 | f | Vanilla |
| 12 | 11 | 83 | m | Strawberry |
| 13 | 12 | 16 | f | Strawberry |
| 14 | 13 | 94 | m | Strawberry |
| 15 | 14 | 55 | m | Vanilla |
| 16 | 15 | 42 | f | Chocolate |
| 17 | 16 | 18 | f | Vanilla |

Raw Data

| Age | |
|---|---|
| Mean | 42.6 |
| Standard Dev. | 21.9 |

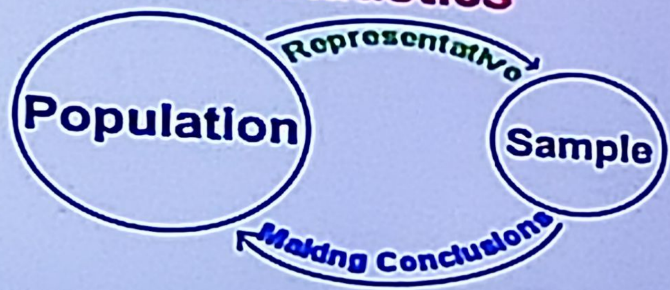


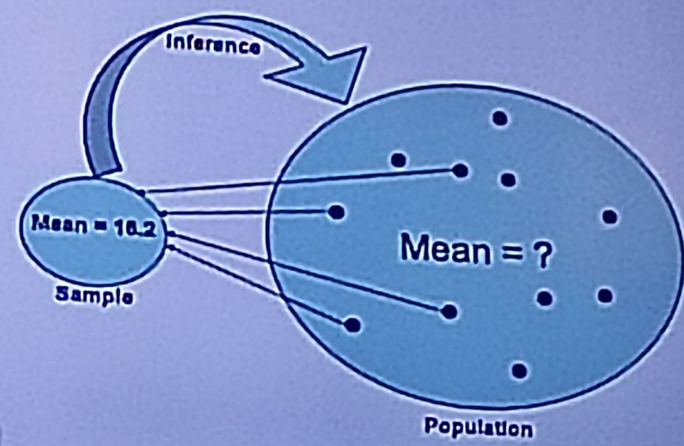

Descriptive Statistics

# Branches of Statistics

3. Inferential statistics: Methods used to analyze a small set of data (sample) to draw a conclusion about a population (generalization).

## Inferential Statistics



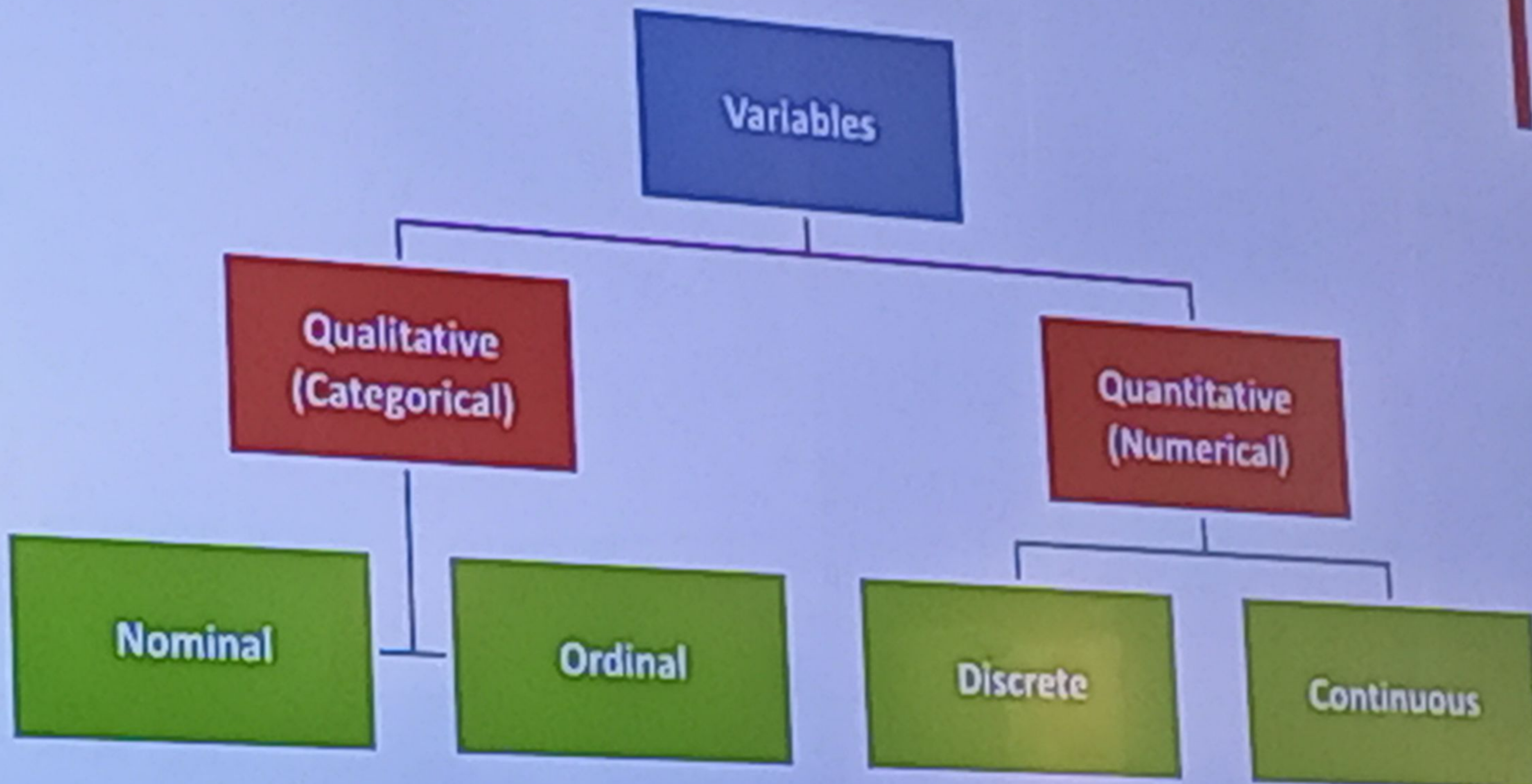
# Variables (traits)

- A variable is a **characteristic** which can take different values in different individuals.
- Examples of variables
  - Bacterial count, shelf life, gender, weight of animal, enzyme activity, color.

# Data

- The **values** of a variable.
- Examples:
  - Birth weights of calves (35.5, 41 .... kg)
  - Litter size of rabbits (1, 2, 3, .... 10)
  - Pain score (1, 2, 3, 4, 5)
  - Gender (male or female)
  - Color of sample (white, yellowish, yellow)

# Types of Variables (Data)



# Qualitative (Categorical) variables

- Values expressed in categories

Examples

- Pain score (no pain, slight pain, …. extreme) (1, 2, … 5)
- Gender (male or female)
- Color of dogs (yellow, brown, black)

# Qualitative (Categorical) variables

## Nominal (Unordered)

- What ice cream flavor do you prefer?

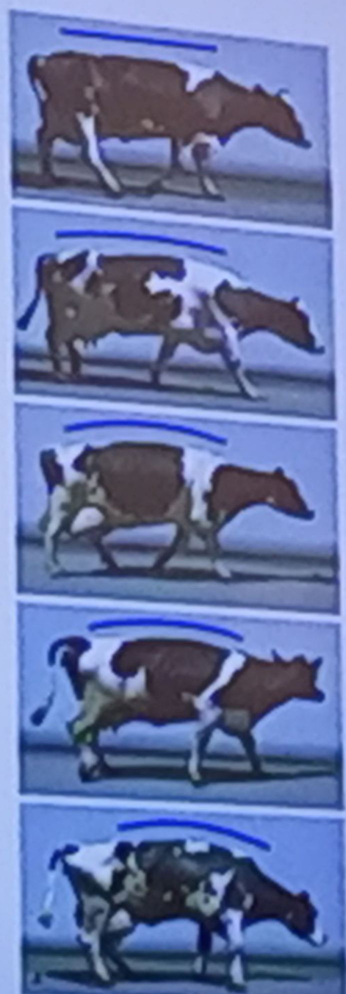
  Vanilla, chocolate, strawberry

## Ordinal (Ordered)

- Pain score

  No pain – slight – moderate – severe – extreme

# Locomotion scoring in cattle



| | | |
|---|---|---|
| 1 | **Normal** | Stands and walks normally with flat back. Long confident strides. |
| 2 | **Mildly Lame** | Stands with flat back, arches when walks. Slightly abnormal gait. |
| 3 | **Moderately Lame** | Stands and walks with arched back. Short strides. |
| 4 | **Lame** | Arched back standing and walking. Favors certain legs. |
| 5 | **Severely Lame** | Constant arched back. Great difficulty moving. |

# Quantitative (numerical) variables

- Values expressed as numbers.
- The differences between values have numerical meaning.

**Discrete**

- They are obtained by counting.
- Only take integer values (1, 2, 6 ....).

Examples

- Litter size
- Number of pregnancies
- Number of egg

# Quantitative (numerical) variables

Continuous

- They are obtained by measuring.

Examples

- Heights
- Weight
- Speed
- Hormone levels.

NB. In SPSS, both discrete and continuous variables are termed "Scale variables"

# Identify the type of variable

- ABO blood groups.
- Egg production per week.
- Body condition score.
- Plasma insulin level.
- Yes or No.
- Services per conception.

# Numerical Summaries of Data

| Descriptive statistics | |
|---|---|
| **Measures of center** | **Measures of variation (dispersion)** |
| - Mean<br>- Median<br>- Mode | - Range<br>- Variance<br>- Standard deviation<br>- Coefficient of variation<br>- IQR |

## Weighted Mean

- Weighted mean is used when some scores contribute more "weight" than others in the final mean.

$$Mw = \frac{\sum(w * x)}{\sum w}$$

- Example: in CGPA calculation

# Weighted Mean

| Subject | Genetics | Biostatistics | Physiology | Anatomy | Biochemistry | Histology | English | Behavior |
|---|---|---|---|---|---|---|---|---|
| GPA | 2.666 | 4.000 | 2.333 | 2.000 | 3.666 | 3.000 | 4.000 | 2.666 |
| CH | 3 | 2 | 2 | 3 | 2 | 3 | 1 | 2 |

$$CGPA = \frac{(2.666 * 3) + (4.000 * 2) + (2.333 * 2) + (2.000 * 3) + (3.666 * 2) + (3.000 * 3) + (4.000 * 1) + (2.666 * 2)}{3 + 2 + 2 + 3 + 2 + 3 + 1 + 2}$$

CGPA = 2.907

# Parameters and Statistics

- **Statistics** are numbers that describe data from a sample
- **Parameters** are numbers that describe data for a population.

- **We use sample statistics to estimate population parameters.** For example, if we use a random sample of lambs, and determine their immune level to a disease, we can calculate the sample mean (**statistic**) and use it as an estimate of the population mean (**parameter**).

| | SAMPLE | POPULATION |
|---|---|---|
| MEAN | $\bar{x}$ | $\mu$ |
| STANDARD DEVIATION | $s$ | $\sigma$ |
| | STATISTIC | PARAMETER |

# Measures of center

They describe the most typical (normal) value in the data (center or middle of data).

- Mean
- Median
- Mode

# The arithmetic mean

The following are milk yield (kg) per day of five goats

2 4 4 3 1

$$\bar{X} = \frac{\sum X_i}{n}$$

$$\bar{X} = \frac{2+4+4+3+1}{5} = 2.8\ kg$$

$$\sum (X_i - \bar{X}) = 0$$

Sorting data from the smallest to the largest

1 2 3 4 4

$n = 5$ (odd number), the median is the value of observation:

$$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{5+1}{2}\right)^{th} = 3^{rd} = 3$$

$$\left(\frac{6}{2}\right)^{th} \; and \; \left(\frac{6}{2}+1\right)^{th} = 3^{rd} and \; 4^{th}$$

The median equals

$$\left(\frac{3+4}{2}\right) = 3.5 \; kg$$

# Mean or Median?

1. The median is a better indicator of the center if the data has outlier(s).

   (An outlier is a value that differs greatly from other values.)

   1 2 3 4 4 5 25

   Mean = 7.29

   Median = 4

2. With ordinal data ( e.g. pain score)

# The mode

- The mode is the value that has the highest frequency in a data set.

2 4 4 3 1

- The mode is 4 kg.